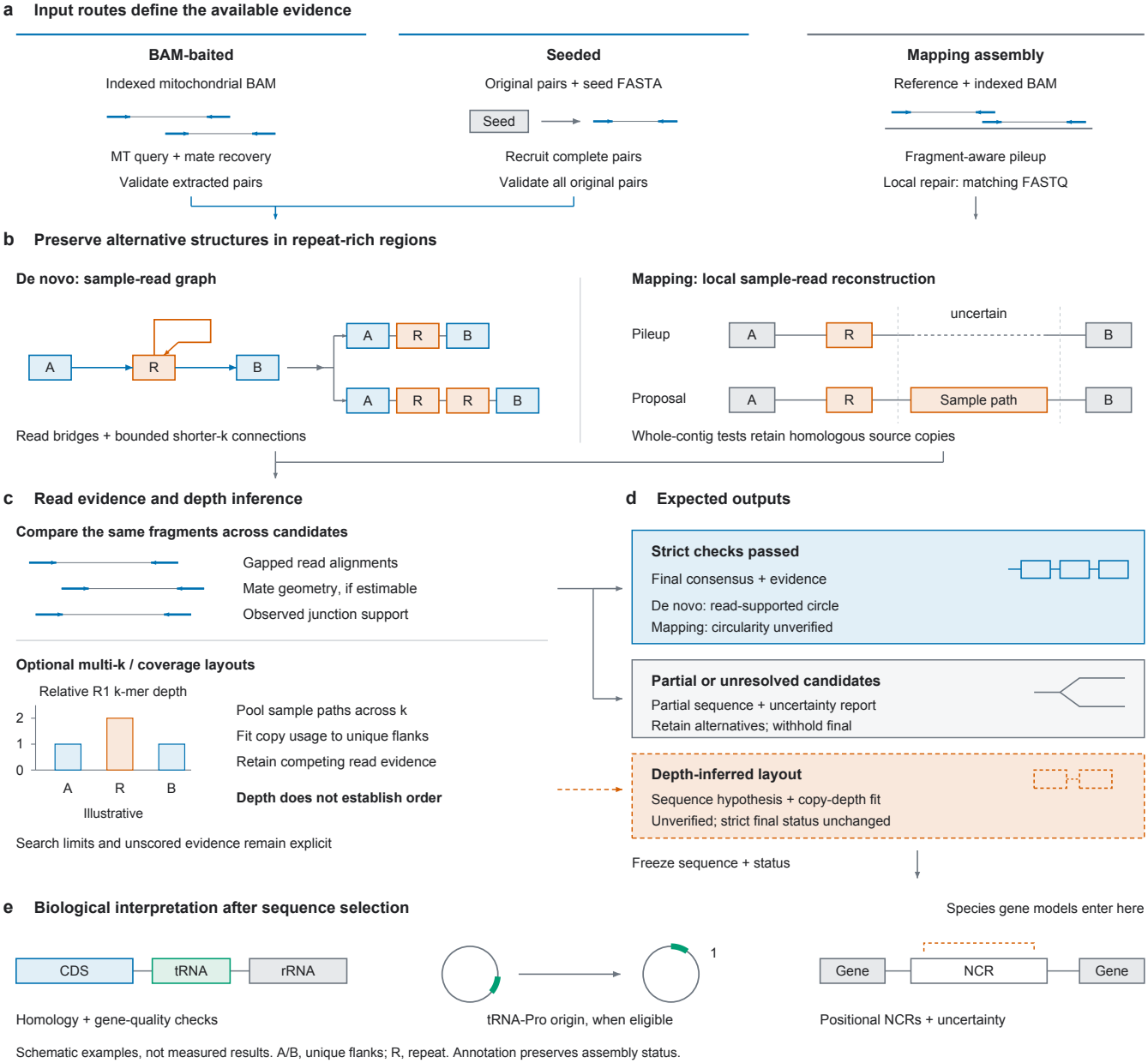

a Input routes define the available evidence
BAM-baited
Indexed mitochondrial BAM
MT query + mate recovery
Validate extracted pairs
Seeded
Original pairs + seed FASTA
Seed
Recruit complete pairs
Validate all original pairs
Mapping assembly
Reference + indexed BAM
Fragment-aware pileup
Local repair: matching FASTQ
b Preserve alternative structures in repeat-rich regions
De novo: sample-read graph
A
R
B
A R B
A R R B
Read bridges + bounded shorter-k connections
Mapping: local sample-read reconstruction
uncertain
Pileup
Proposal
Sample path
Whole-contig tests retain homologous source copies
c Read evidence and depth inference
Compare the same fragments across candidates
Gapped read alignments
Mate geometry, if estimable
Observed junction support
Optional multi-k / coverage layouts
Relative R1 k-mer depth
2
1
0
Illustrative
Pool sample paths across k
Fit copy usage to unique flanks
Retain competing read evidence
Depth does not establish order
Search limits and unscored evidence remain explicit
d Expected outputs
Strict checks passed
Final consensus + evidence
De novo: read-supported circle
Mapping: circularity unverified
Partial or unresolved candidates
Partial sequence + uncertainty report
Retain alternatives; withhold final
Depth-inferred layout
Sequence hypothesis + copy-depth fit
Unverified; strict final status unchanged
Freeze sequence + status
e Biological interpretation after sequence selection
Species gene models enter here
CDS
tRNA
rRNA
1
Gene
NCR
Gene
Homology + gene-quality checks
tRNA-Pro origin, when eligible
Positional NCRs + uncertainty
Schematic examples, not measured results. A/B, unique flanks; R, repeat. Annotation preserves assembly status.